

CALCULATED FIELDS WITH DAX

CALCULATE

CALCULATE()

Evaluates an expression in a context that is modified by filters

=**CALCULATE**(Expression, [Filter1], [Filter2],...)

Name of an **existing measure** or a **DAX formula** for a valid measure

Examples:

- [Total Orders]
- SUM('Returns Data'[Return Quantity])

A Boolean (True/False) expression or a table expression that defines a filter

Note: these require fixed values or aggregation functions that return a scalar value (you cannot create filters based on measures)

Examples:

- 'Territory Lookup'[Country] = "USA"
- Calendar[Year] <> MAX(Calendar[Year])



PRO TIP:

Think of CALCULATE as a **filter modifier**; it allows you to overrule existing report filters and "force" new filter context

EXAMPLE: CALCULATE

1 Red Sales = CALCULATE([Quantity Sold], 'Product Lookup'[Product Color] = "Red")

Here we've defined a new measure named **Red Sales**, which evaluates the **Quantity Sold** measure under a filter context where the product color is "Red"

Product Color	Quantity Sold	Red Sales
Black	10,590	4,011
Multi	5,756	4,011
Red	4,011	4,011
Silver	3,257	4,011
Total	23,614	4,011

Note how we see the **the same repeated values** for each product color, and even the total!

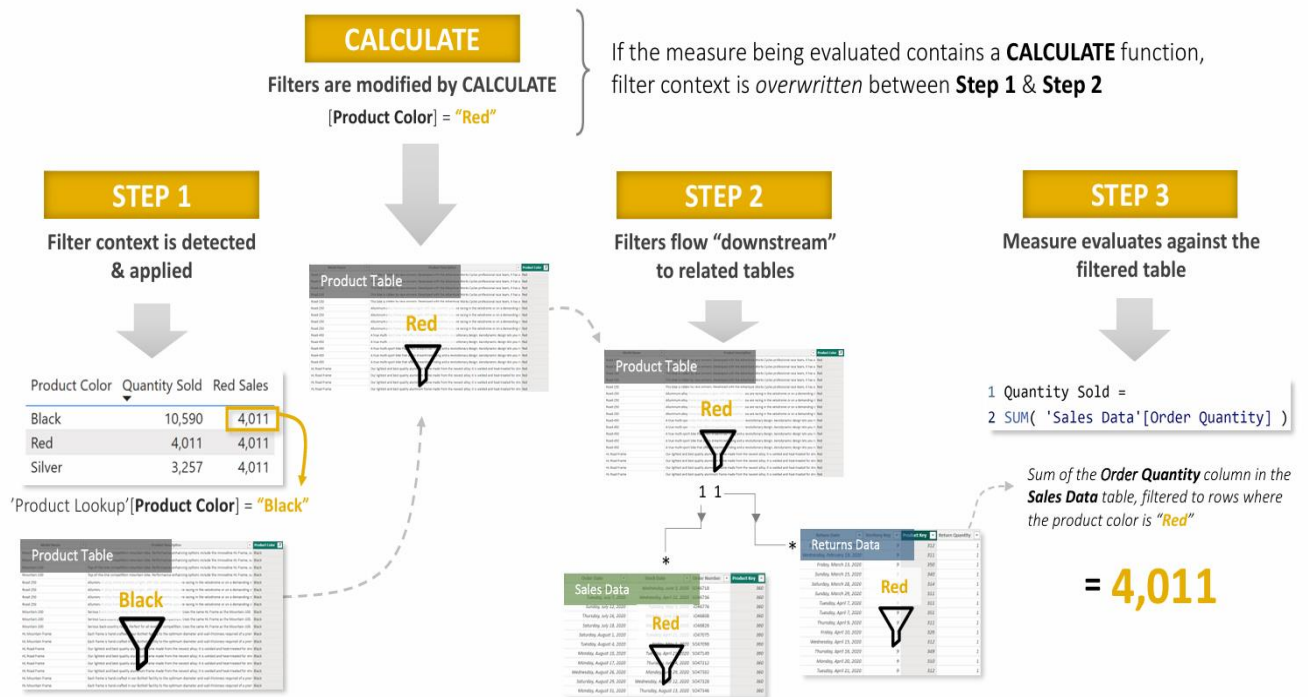


HEY THIS IS IMPORTANT!

The CALCULATE function modifies and overrules any competing filter context!

In this matrix, the "Black" row has competing filter context: Product Color = **Black** (from the row label) and Product Color = "Red" (from the CALCULATE function)

Both can't be true at the same time, so the "Red" filter from CALCULATE takes priority



```

1 Bike Returns =
2 CALCULATE(
3     [Total Returns],
4     'Product Categories Lookup'[Category Name] = "Bikes"
5 )

```

ALL

ALL

Returns all rows in a table, or all values in a column, ignoring any filters that have been applied

=ALL(Table or Column, [Column2], [Column3],...)

The **table** or **column** that you want to clear filters on

Examples:

- Transactions
- Products[Category]

Additional columns that you want to clear filters on (optional)

- Cannot specify columns if your first parameter is a **table**
- All columns must include the **table name** and come from the **same table**

Examples:

- 'Customer Lookup'[City], 'Customer Lookup'[Country]
- Products[Product Name]



PRO TIP:

Instead of adding filter context, the **ALL** function removes it. This is often used in "% of Total" calculations, when the denominator needs to remain fixed regardless of filter context.

```

1 All Returns =
2 CALCULATE(
3     [Total Returns],
4     ALL(
5         'Returns Data'
6     )
7 )

```

```

1 % of All Returns =
2 DIVIDE(
3     [Total Returns],
4     [All Returns]
5 )

```

Category Name	% of All Returns
Bikes	23.60%
Clothing	14.76%
Accessories	61.64%
Total	100.00%

TIME INTELLIGENCE

Performance
To-Date

=**CALCULATE**(Measure, **DATESYTD**(Calendar[Date]))

Use **DATESYTD** for Years, **DATESQTD** for Quarters, **DATESMTD** for Months

Previous
Period

=**CALCULATE**(Measure, **DATEADD**(Calendar[Date], -1, **MONTH**))

Select an interval (**DAY**, **MONTH**, **QUARTER**, or **YEAR**) and the # of intervals to compare (e.g. previous month, rolling 10-day)

Running
Total

=**CALCULATE**(Measure, **DATESINPERIOD**(Calendar[Date], **MAX**(Calendar[Date]), -10, **DAY**))



PRO TIP:

To calculate a **moving average**, use the running total calculation above and **divide by the number of intervals**

```

1 Previous Month Orders =
2 CALCULATE(
3     [Total Orders],
4     DATEADD(
5         'Calendar Lookup'[Date],
6         -1,
7         MONTH
8     )
9 )

```

```

1 90-day Rolling Profit =
2 CALCULATE(
3     [Total Profit],
4     DATESINPERIOD(
5         'Calendar Lookup'[Date],
6         LASTDATE(
7             'Calendar Lookup'[Date]
8         ),
9         -90,
10        DAY
11    )
12 )

```

```

1 Order Target =
2 [Previous Month Orders] * 1.1

```